

**Mass dimension: 6**

**nD: 0, H: 2, H+: 1, Q+: 1, dC+: 1**

**SU3\_C-Basis Tensors**

$$(T_{SU3})_{abc}^0 = +\left(\frac{1}{2}\right)\varepsilon_{abc} - \left(\frac{1}{2}\right)\varepsilon_{bac} \quad (1)$$

**SU2\_W-Basis Tensors**

$$(T_{SU2})_{ijkl}^0 = \varepsilon_{ik}\varepsilon_{jl} \quad (2)$$

$$(T_{SU2})_{ijkl}^1 = \varepsilon_{ij}\varepsilon_{kl} \quad (3)$$

**H0H0H+0Q+0dC+0**

$$+\left(\frac{1}{2}i\right)M_s^{-2}(y_d)^{wu}(y_d)^{vx}(y_d^\dagger)^{uv}\varepsilon_{\dot{\alpha}_2\dot{\alpha}_1}(T_{SU2})_{ijkl}^0(T_{SU3})_{abc}^0H_iH_j(H^\dagger)_k(Q^\dagger)_{l,ab}^{\dot{\alpha}_1,w}(d_{\mathbb{C}}^\dagger)_c^{\dot{\alpha}_2,x} \quad (4)$$

$$-\left(\frac{1}{4}\right)M_s^{-2}(y_d)^{wu}(y_d)^{vx}(y_d^\dagger)^{uv}\varepsilon_{\dot{\alpha}_2\dot{\alpha}_1}(T_{SU2})_{ijkl}^1(T_{SU3})_{abc}^0H_iH_j(H^\dagger)_k(Q^\dagger)_{l,ab}^{\dot{\alpha}_1,w}(d_{\mathbb{C}}^\dagger)_c^{\dot{\alpha}_2,x} \quad (5)$$

**nD: 1, Q: 1, H: 1, H+: 1, Q+: 1**

**SU3\_C-Basis Tensors**

$$(T_{SU3})_{abc}^0 = +\left(\frac{1}{2}\right)\varepsilon_{abc} - \left(\frac{1}{2}\right)\varepsilon_{acb} \quad (6)$$

**SU2\_W-Basis Tensors**

$$(T_{SU2})_{ijkl}^0 = \varepsilon_{ik}\varepsilon_{jl} \quad (7)$$

$$(T_{SU2})_{ijkl}^1 = \varepsilon_{ij}\varepsilon_{kl} \quad (8)$$

**Q0H0H+1Q+0**

$$+\left(\frac{1}{2}\right)M_s^{-2}(y_d)^{wu}(y_d^\dagger)^{uv}\varepsilon^{\alpha_1\alpha_2}\varepsilon_{\dot{\alpha}_1\dot{\alpha}_2}(T_{SU2})_{ijkl}^0(T_{SU3})_{abc}^0Q_{\alpha_1,i,a}^vH_jD_{\alpha_2}^{\dot{\alpha}_1}(H^\dagger)_k(Q^\dagger)_{l,bc}^{\dot{\alpha}_2,w} \quad (9)$$

**nD: 0, Q: 1, dC: 1, H: 1, H+: 2**

**SU3\_C-Basis Tensors**

$$(T_{SU3})_{abc}^0 = +\left(\frac{1}{2}\right)\varepsilon_{abc} - \left(\frac{1}{2}\right)\varepsilon_{acb} \quad (10)$$

**SU2\_W-Basis Tensors**

$$(T_{SU2})_{ijkl}^0 = \varepsilon_{ik}\varepsilon_{jl} \quad (11)$$

$$(T_{SU2})_{ijkl}^1 = \varepsilon_{ij}\varepsilon_{kl} \quad (12)$$

**Q0dC0H0H+0H+0**

$$-\left(\frac{1}{2}i\right)M_s^{-2}(y_d)^{uv}(y_d^\dagger)^{xu}(y_d^\dagger)^{vw}\varepsilon^{\alpha_2\alpha_1}(T_{SU2})_{ijkl}^0(T_{SU3})_{abc}^0Q_{\alpha_1,i,a}^w(d_{\mathbb{C}})_{\alpha_2,bc}^xH_j(H^\dagger)_k(H^\dagger)_l \quad (13)$$

$$+\left(\frac{1}{4}\right)M_s^{-2}(y_d)^{uv}(y_d^\dagger)^{xu}(y_d^\dagger)^{vw}\varepsilon^{\alpha_2\alpha_1}(T_{SU2})_{ijkl}^1(T_{SU3})_{abc}^0Q_{\alpha_1,i,a}^w(d_{\mathbb{C}})_{\alpha_2,bc}^xH_j(H^\dagger)_k(H^\dagger)_l \quad (14)$$